Cell Tuition
1-1 PRIVATE TUTORS

ALGEBRA

Linear Inequalities

Name _____ Date _____ No calculator Version 1 Scan to practise this topic online

KEY WORDS inequality *(whārite-kore)* · greater than *(rahi ake i)* · less than *(iti iho i)* · at least · at most

A Reading and writing inequalities ●○○

Remember: $<$ less than, $>$ greater than, $\leq$ less than or equal to, $\geq$ greater than or equal to.

1. Write each sentence as an inequality.

a) x is greater than 5 _____

b) x is at most 7 _____

c) x is at least −2 _____

d) x is below 10 _____

e) x is no more than 4 _____

f) x is more than −3 _____

2. Circle every number that makes the inequality true.

a) $x > 3$: 1, 3, 3.5, 8

b) $x \leq -2$: -5, -2, 0, 2

c) $x \geq 0$: -1, 0, 0.1, 6

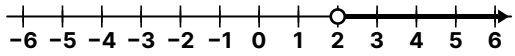
d) $x < -4$: -7, -4.5, -4, 4

B Number lines ●○○

Remember: An open circle means the number is not included. A filled circle means it is included.

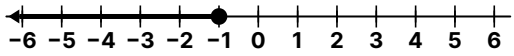
3. Write the inequality shown on each number line.

a)



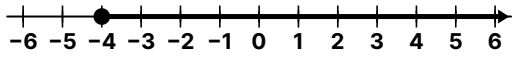
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b)



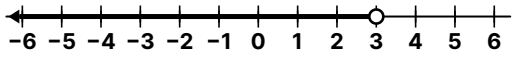
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c)



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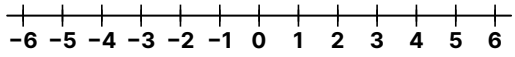
d)



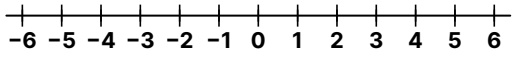
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4. Show each inequality on its number line.

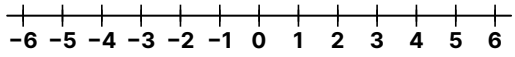
a) $x \geq 1$



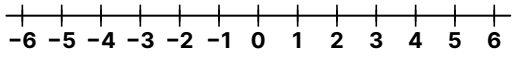
b) $x < -2$



c) $x > -5$



d) $x \leq 4$



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C Solving inequalities ●●○

Remember: Solve an inequality the same way as an equation. Keep the inequality sign in every line.

5. Solve.

a) $x + 3 > 7$ **b)** $2x \leq 10$ **c)** $x - 6 < -8$

d) $3x - 1 \geq 11$ **e)** $5x + 2 > 17$ **f)** $4x - 7 \leq 9$

g) $\frac{x}{2} < 4$ **h)** $2x + 9 \geq 3$ **i)** $\frac{x}{3} + 1 > 4$

j) $2(x + 3) > 10$ **k)** $3(x - 2) \leq 9$ **l)** $5(2x + 1) < 15$

D Dividing by a negative ●●●

Watch out: When you multiply or divide both sides by a negative number, turn the inequality sign around.

6. Solve.

a) $-2x > 6$ **b)** $-x \leq -4$ **c)** $-3x + 1 < 10$

d) $5 - x \leq 2$ **e)** $10 - 2x > 4$ **f)** $-4x - 3 \geq 5$

g) $7 - 3x < 13$ **h)** $-5x + 2 \leq -8$ **i)** $1 - 4x \geq 13$

E Both sides and fractions ●●●

7. Solve.

a) $5x - 3 \geq 2x + 9$ **b)** $7x + 2 < 3x - 6$ **c)** $4x + 5 > x + 14$

d) $2x + 9 \leq 5x + 3$ **e)** $\frac{x + 1}{2} < 4$ **f)** $\frac{2x - 1}{3} \geq 3$

g) $\frac{3x + 4}{2} > 5$ **h)** $3x - 8 > 6x + 4$ **i)** $\frac{5x - 2}{4} \leq 7$

★ Extension

8. Solve $-3 < 2x + 1 \leq 9$. Then list the whole numbers that satisfy it.

9. Movie tickets cost \$14 each and a large popcorn costs \$8. Mere has \$50 and buys one popcorn. Write an inequality for the number of tickets x she can buy, and solve it. What is the largest number of tickets?

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Linear Inequalities

Name _____ Date _____ No calculator Version 2 Scan to practise this topic online

KEY WORDS inequality *(whārite-kore)* · greater than *(rahi ake i)* · less than *(iti iho i)* · at least · at most

A Reading and writing inequalities ●○○

Remember: $<$ less than, $>$ greater than, $\leq$ less than or equal to, $\geq$ greater than or equal to.

1. Write each sentence as an inequality.

a) x is greater than 8 _____

b) x is at most 3 _____

c) x is at least -4 _____

d) x is below 6 _____

e) x is no more than 9 _____

f) x is more than -1 _____

2. Circle every number that makes the inequality true.

a) $x > 2$: 0, 2, 2.5, 9

b) $x \leq -3$: -6, -3, 0, 3

c) $x \geq 1$: -1, 0.9, 1, 4

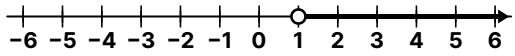
d) $x < -1$: -3, -1.5, -1, 1

B Number lines ●○○

Remember: An open circle means the number is not included. A filled circle means it is included.

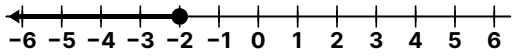
3. Write the inequality shown on each number line.

a)



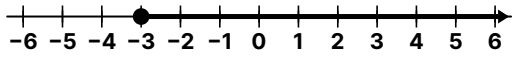
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b)



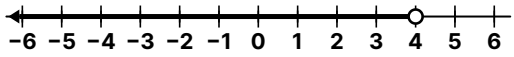
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c)



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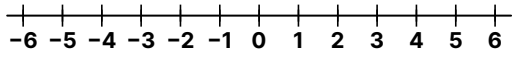
d)



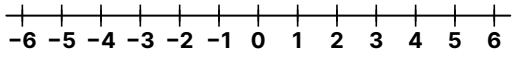
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4. Show each inequality on its number line.

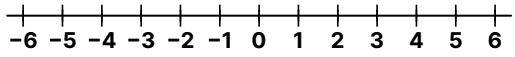
a) $x \geq 2$



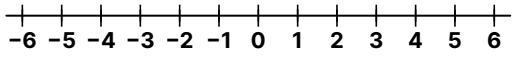
b) $x < -1$



c) $x > -4$



d) $x \leq 5$



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C Solving inequalities ●●○

Remember: Solve an inequality the same way as an equation. Keep the inequality sign in every line.

5. Solve.

a) $x + 5 > 8$ **b)** $3x \leq 12$ **c)** $x - 4 < -7$

d) $2x - 3 \geq 7$ **e)** $4x + 3 > 11$ **f)** $5x - 6 \leq 9$

g) $\frac{x}{3} < 4$ **h)** $2x + 11 \geq 3$ **i)** $\frac{x}{4} + 2 > 4$

j) $3(x + 2) > 12$ **k)** $2(x - 5) \leq 6$ **l)** $4(2x + 1) < 12$

D Dividing by a negative ●●●

Watch out: When you multiply or divide both sides by a negative number, turn the inequality sign around.

6. Solve.

a) $-3x > 6$ **b)** $-x \leq -6$ **c)** $-2x + 3 < 11$

d) $8 - x \leq 3$ **e)** $12 - 3x > 6$ **f)** $-5x - 2 \geq 8$

g) $9 - 2x < 15$ **h)** $-4x + 1 \leq -11$ **i)** $2 - 5x \geq 12$

E Both sides and fractions ●●●

7. Solve.

a) $6x - 5 \geq 2x + 7$ **b)** $8x + 3 < 5x - 6$ **c)** $5x + 4 > 2x + 10$

d) $3x + 11 \leq 7x + 3$ **e)** $\frac{x + 3}{2} < 6$ **f)** $\frac{2x - 3}{5} \geq 3$

g) $\frac{3x + 1}{2} > 5$ **h)** $2x - 9 > 5x + 6$ **i)** $\frac{4x - 2}{3} \leq 6$

★ Extension

8. Solve $-1 < 3x + 2 \leq 11$. Then list the whole numbers that satisfy it.

9. Movie tickets cost \$16 each and a large popcorn costs \$6. Hemi has \$70 and buys one popcorn. Write an inequality for the number of tickets x he can buy, and solve it. What is the largest number of tickets?

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